

CoRoot: A Collaborative Planting System to Strengthen Intergenerational Relationships

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ABSTRACT

The grandparent-grandchild relationship plays an essential and irreplaceable role in life. Strong intergenerational relationships benefit both grandparents and grandchildren. However, social conditions and geographical isolation make it difficult for young people to develop close relationships with their grandparents. In this paper, we explore the collaborative planting system, CoRoot, which is designed to facilitate intergenerational relationships. CoRoot allows young people and their grandparents to water each other's vegetables at a distance and to harvest ripe vegetables through collaborative watering. This paper presents the system design of CoRoot, the interactive mode of the system, and a user study of its usability. We found that collaborative planting can create a link between generations. Our design is expected to provide opportunities for intergenerational communication and promote the emotional bond between grandparents and young people.

CCS CONCEPTS

• **Human-centered computing** → *Collaborative interaction*.

KEYWORDS

Intergenerational relationship, grow vegetables, collaborative watering, communication, connection

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1 INTRODUCTION

In recent years, social, cultural, and demographic changes have contributed to an increased involvement of grandparenting in rearing the next generation [4]. However, grandparents and grandchildren often find it difficult to maintain close relationships due to geographical distances and other social circumstances [1]. Although various technologies have been presented to support intergenerational communication over distance, conversations over messages

or simple phone calls are not as rich and are insufficient for sharing information and establishing strong emotional bonds between grandparents and grandchildren [2, 6].

To strengthen intergenerational relationships, scholars have explored a variety of avenues. In a study conducted by Geurts et al., it was found that more frequent contact with grandparents in childhood helps to strengthen their lifelong relationships [4]. Azadeh Forghani's research suggests that although parents play an influential role in grandparent-grandchild communication, creating an independent and sincere way of communication between grandparents and grandchildren has a positive effect on intergenerational relationships [5]. According to research findings, family, education, leisure, and friendship are the most frequently discussed topics between grandparents and grandchildren [16]. Bente Evjemo's study further suggests that intergenerational communication is closely tied to concurrent activities [1]. The findings of these studies provide theoretical guidance for design practice. Studies have explored a number of ways to promote the relationship between grandparents and young children, but few studies have examined the communication problems between grandparents and young people.

In this paper, we explore CoRoot, a collaborative planting system designed to facilitate intergenerational relationships (Figure 1). CoRoot allows young people and their grandparents to water each other's vegetables at a distance and to harvest ripe vegetables through collaborative watering. The CoRoot system consists of two parts: hardware and software (Figure 2). The hardware includes two smart watering devices and two ladles. The software provides the functions of watering, recording and reviewing. Our design is expected to provide opportunities for intergenerational communication through collaborative planting and foster the emotional bond between grandparents and young people.

Next, we conducted a field evaluation of CoRoot. We evaluated our system with a family for six weeks and we focused on:

- Whether the system can effectively promote intergenerational communication?
- What design factors are effective at facilitating intergenerational communication?

As a result of our research, we believe that CoRoot can promote contact and communication between grandparents and young people. For grandparents, collaborative planting is not only a hobby in their spare time but also a way to realize personal value by sharing vegetables with friends and family members. For young people, growing vegetables is always a pleasure in their busy life without taking up too much time. Collaborative planting brings people closer together, and both grandparents and grandchildren gain a

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great deal of enjoyment from it. CoRoot will become an engaging and interactive medium for grandparents and young people.



Figure 1: CoRoots allows grandparents and grandchildren to grow vegetables through collaborative watering.



Figure 2: There are two parts to the CoRoot system: hardware and software.

2 RELATED WORK

Several studies have demonstrated the importance and irreplaceability of the relationship between grandparents and grandchildren [7, 17]. Strong intergenerational relationships are beneficial to grandparents and grandchildren [7]. With high levels of stress in today's society and few ways to cope, young people often turn to their grandparents for emotional support [7, 15]. However, maintaining

close intergenerational relationships is a great challenge for many people. Based on literature research, we determined that there are three main factors affecting intergenerational relationships:

- As a result of parental intervention, communication between grandparents and grandchildren is often forced rather than genuine. As Becker et al. indicated, parents are often the “driving force” for fostering grandparent-grandchild communication, and both grandparents and young people have problems maintaining their relationships independently [2, 6].
- Due to geographical separation and other social circumstances, young people find it difficult to establish intimate relationships with their grandparents [8, 9].
- Since the cognitive and technological gap between the two generations is enormous, phone calls are the most common choice for communication between grandparents and young people [11], which proves insufficient for sharing information and building a closer connection [6, 11].

In response to the issues described above, a variety of communication systems have been proposed to support intergenerational relationships. Take Me With You was designed to promote grandparent-grandchild interactions through gamification [9]. People in Books was an application developed by Follmer et al. to facilitate remote storytelling [14]. Similarly, StoryBox served as a medium for sharing photos and handcrafts among family members [17]. While these systems might help young children to keep in touch with their grandparents, few studies have provided insights into the communication problems between young people and their grandparents.

For more insights, we conducted online or face-to-face semi-structured interviews exploring the communication patterns and challenges people face in maintaining relationships between young people and their grandparents. For the interviews, we recruited four young adults (aged 18-35) and three elders (aged 65-78). Participants were from different backgrounds and did not live with their grandparents or grandchildren. Throughout the interviews, it was no surprise to discover that each participant had their own unique communication need, but all expressed a strong desire to stay in touch with their distant grandparents or grandchildren. We synthesize the insights from the interviews and find that:

For young people:

- Due to their busy schedules, they hardly have time to communicate with their grandparents.
- Meeting their grandparents occasionally is a heartwarming experience.
- They hope to stay connected with their grandparents by doing something together that is not time-consuming.

For older adults:

- While they want to stay in touch with their adult grandchildren, they do not want to bother them too much.
- During their free time, they enjoy planting.
- By sharing with others, they gain a sense of accomplishment.

We captured quotes from participants during the interviews:

“Calling my grandmother is a rule for me, but sometimes I feel ashamed.”

“My grandmother likes growing vegetables, and I miss my grandmother’s cooking most. It is enjoyable to have dinner with grandmother at her house because there is a pleasant family atmosphere that I yearn for.”

Based on the interview findings, people neglect to maintain frequent contact with their distant grandparents or grandchildren due to the existing bonds of kinship. Participants found that conversations over messages or simple phone calls were insufficient for sharing information and hoped to stay connected by doing something together. We suggested that “vegetable growing” could serve as a medium for connecting grandparents with young people, and all participants were enthusiastic about the idea.

3 COROOT SYSTEM

The CoRoot system is designed to facilitate intergenerational relationships through collaborative planting. There are two parts to the CoRoot system: hardware and software (Figure 2). The hardware includes two smart watering devices and two ladles. The software provides the functions of watering, recording and reviewing. With this system, each participant can water their vegetables with a ladle (the vegetable contains half of the water necessary now) and then help each other finish the rest of the watering over a distance using a smart watering device or app. In CoRoot, grandparents and young people work together to grow vegetables.

Design goals

- Young people and grandparents collaborate on watering vegetables.
- Ensure that older adults can operate in an easy and traditional manner.
- For young people, provide an interactive mode that is convenient and time-saving.

User journey (Figure 3)

- Plant the seeds in two flowerpots and water them daily by grandmother.
- During the early stages of planting, grandmother waters two pots of vegetables and enjoys the fun.
- After the vegetables reach the seedling stage, the grandchild takes home a pot of vegetables to continue growing.
- Grandmother and grandchild grow vegetables together with CoRoot.
- When the vegetables are ready to eat, they can share them with family members.

Collaborative watering via CoRoot (Figure 4)

Watering grandmother’s vegetables:

- Grandmother waters her vegetables with a ladle (the vegetable contains half of the water necessary now). After that, she uses her smart watering device to send a message to remind her grandchild to help her water.
- The watering information will be synchronized to the grandchild’s mobile phone and the smart watering device.
- Grandchild can assist their grandmother with the rest of the watering remotely by using the app or a smart watering device.

- Finish watering. Grandmother can express her gratitude to her grandchild.

Watering grandchild’s vegetables:

The process of watering grandchild’s vegetables is the same as that of watering grandmother’s.



Figure 3: User journey of growing vegetables collaboratively.

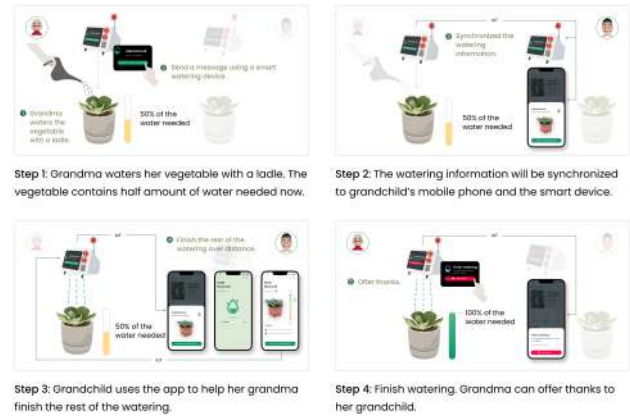


Figure 4: Collaborate on watering vegetables via CoRoot.

Software application

We designed a software application that allows people to grow vegetables collaboratively from a distance. The app consists of 3 modules:

- **Vegetable watering.** In this part, users can (a) make water plans; (b) remind each other to water; (c) check water records.
- **Vegetable growth recording.** In this part, users can (a) review the growing process of vegetables; (b) view live photos of vegetables.
- **Reviewing.** Users can make memorial videos with their photos.

Apart from watering and recording vegetable growth, the app generates vegetable-growing photos based on the database and automatically makes memorial videos. It creates a powerful emotional connection between grandparents and young people.

4 USER STUDIES

To test the usability of CoRoot, we evaluated our system with a family (grandmother aged 75, grandchild aged 21, and participants lived apart) for six weeks, focusing on:

- Whether the system can effectively promote intergenerational communication?
- What design factors are effective at facilitating intergenerational communication?

As part of the pre-evaluation process, we surveyed participants regarding their existing intergenerational relationships, communication frequency, channel, and content they had with their distant grandparents or grandchildren. Then, we set up CoRoot and explained its features to participants. For the next six weeks, participants would collaborate on growing vegetables via CoRoot. The experimental material we used for this study was kale broccoli seeds. At the end of the study, we conducted interviews with the participants to discuss how CoRoot influenced communication patterns between grandparents and young people. Furthermore, we examined users' overall experience with the system. Following are the vegetables' growing processes through collaborative watering via CoRoot (Figure 5).



Figure 5: Through collaborative watering, grandmother's (the left) and grandchild's vegetables (the right) grow.

Results

Result 1. To identify whether the system can promote intergenerational communication, we analyzed the data using Biemans et al.'s established coding categories [3] that were used to study social connectedness between friends and family members. We analyzed the frequency and types of conversations between grandmother and grandchild on WeChat (Table 1 and Table 2). Following the study, we investigated the frequency of phone calls between generations before and during the study (Table 3) and conducted interviews with the participants. We found that phone calls and WeChat communications between grandmother and grandchild significantly increased. While grandmother and grandchild still communicated by telephone or WeChat, the content of their conversations became more affluent and more diverse over time. "Whenever I called my

grandma, it seemed to be common greetings. However, growing vegetables with grandmother offered me an opportunity to chat with her, which was a heartwarming experience." The results show that CoRoot effectively promotes intergenerational communication and creates a stronger emotional bond between grandparents and young people.

Result 2. We interviewed participants to understand what design factors are effective at facilitating intergenerational communication. We found that collaborative planting offered grandparents and young people opportunities to interact and communicate. Participants needed to remind, chat, and share to accomplish the joint goal of collaborative watering, which helped foster communication. Further, CoRoot preserves the traditional way of watering for older adults while providing a convenient and time-saving interactive mode for young people. For grandparents, collaborative planting is not only a hobby in their spare time but also a way to realize personal value by sharing vegetables with friends and family members. For young people, growing vegetables in their busy lives is always a joy without taking up too much time. "City born and city bred, I regarded planting as a tiring business. Growing vegetables with my grandma and feeling the power of love and companionship, the inspiration and healing sense gained from collaborative planting is a memorable experience that provides the human touch I yearn for." Collaborative planting brings people closer together, and both grandparents and grandchildren gain a great deal of enjoyment from it.

5 DISCUSSION AND FUTURE WORK

The relationship between generations is unique in life. Encouraging communication between young people and their grandparents can not only make older adults feel loved and warm but also help young people learn to share, cherish and appreciate. In this paper, we designed CoRoot aimed at facilitating intergenerational communication. CoRoot allows users to grow vegetables through collaborative watering. The results of our research indicate that collaborative planting can enhance intergenerational communication and foster a closer emotional bond between young people and their grandparents.

O'Daniel M's theory suggests that collaboration creates cohesion and promotes effective communication [13]. Using CoRoot to collaborate on growing vegetables makes the interaction between generations more enjoyable and allows for more communication. Collaborating on growing vegetables makes frequent communication possible, and frequent contact increases a more profound understanding between grandparents and young people. Collaboration brings people closer together, and the two generations establish more robust connections and harvest heartwarming memories during collaborative watering. The importance of collaboration and shared experiences in maintaining intergenerational relationships is highlighted in our research. Collaboration by itself supports connections and a feeling of belonging, and shared experience with another person can also provide pathways towards connections: through self-other overlap and through emotional cohesion and empathy. CoRoot provides opportunities to collaborate and creates shared experience for participants, and it proved to be an engaging and interactive medium for grandparents and young people.

Table 1: Number and types of threads shared between grandmother and grandchild via WeChat before the study

Before study	Total number of messages	Greetings	Everyday life	Special events	Something funny
Grandmother	13	4	7	1	1
Grandchild	12	7	3	2	0

Table 2: Number and types of threads shared between grandmother and grandchild via WeChat during the study

During study	Total number of messages	Greetings	Everyday life	Special events	Something funny
Grandmother	151	26	96	15	14
Grandchild	159	49	77	6	27

Table 3: Frequency of phone calls between grandmother and grandchild

Time and behavior	Before the study	During the study
Duration	42 days	42 days
Grandmother calling grandchild	4	16
Grandchild calling grandmother	6	20

We also explore strategies for promoting intergenerational interaction in our study. During our research, participants engaged in "collaborative making". As we saw in previous findings, the intergenerational setting often meant tellers were unaware of who their listeners were. This uncertainty could result in a "teller's block" of sorts, when tellers are unable to sufficiently imagine the future audience with whom they wish to share[10]. Young people and older adults are passive teller or listener in this context. Jeffrey K. described intimate family relationships as "a negotiated process of sense-making"[12]. In our study, collaborative making is regarded as a negotiated process that strengthens relationships between generations. For example, grandparents and young people need to negotiate watering tasks and share the vegetable growing process weekly during collaborative watering. Participants can become active seekers in the collaborative watering process in order to achieve their common goals. We use this experience to draw conclusions about "collaborative making" as a gateway to mutually engaging communication, and it can be a pathway to enhance intergenerational interaction. In the process of collaborative making, older adults and young people are in a more proactive role, which can stimulate a real sense of connection and promote intergenerational relationships.

However, CoRoot still has some limitations, which we will have to address in the future. Though the system has received positive feedback from participants, it is currently being tested in a family for six weeks. This research will continue to explore the use of the system in a larger number of families over a more extended period. In addition, we also displayed our design to HCI experts and received valuable suggestions to be included in our next iteration.

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